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SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 1

MICROSOFT AZURE TUTORIAL REPORT

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GROUP 3

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TABLE OF CONTENTS

Project Overview	3
Description on the Solution	4
Technology Use	6
Data Pipeline Architecture	8
Microsoft Power BI Dashboard	12
Reflection	15

Project Overview

This project demonstrates the design and implementation of an end-to-end data pipeline using Microsoft Azure services. The primary objective of this project is to build a scalable and automated workflow that ingests raw data into a data lake, transforms the data into a structured format, and identifies valuable insights through data visualization tools.

In modern businesses, organizations rely heavily on building efficient data pipelines to process massive amounts of data from multiple sources. This project simulates a real-world business scenario to extract customer and sales data from an on-premises SQL database, transform the data into the cloud, and generate meaningful insights through a Power BI dashboard.

The main objectives of the project are:

- To design an automated data pipeline using Azure services
- To ingest data from source systems into cloud storage
- To transform raw data into structured and analysis-ready data
- To store processed data for efficient querying
- To visualize insights using a business intelligence tool

Overall, this project focuses on batch processing using Microsoft Azure services, demonstrating real-world data engineering practices.

Description of the Solution

The proposed solution implements a well-built, automated ETL (Extract, Transform, Load) workflow designed to reduce the gap between static on-premises data and dynamic cloud-based analytics. The Medallion Architecture is employed in this solution and uses multi-stage refinement processes to ensure high data quality.

The logical workflow consists of the following components:

Data Ingestion:

The initial step involves capturing data on customer and sales from an on-premises SQL database. Through the Ingestion Runtime, Azure Data Factory (ADF) acts as the orchestration engine in a safe way, ingesting the raw data into Azure Data Lake Storage (ADLS) Gen 2.

Data Transformation (Medallion Architecture):

The data is processed using Azure Databricks in three different layers:

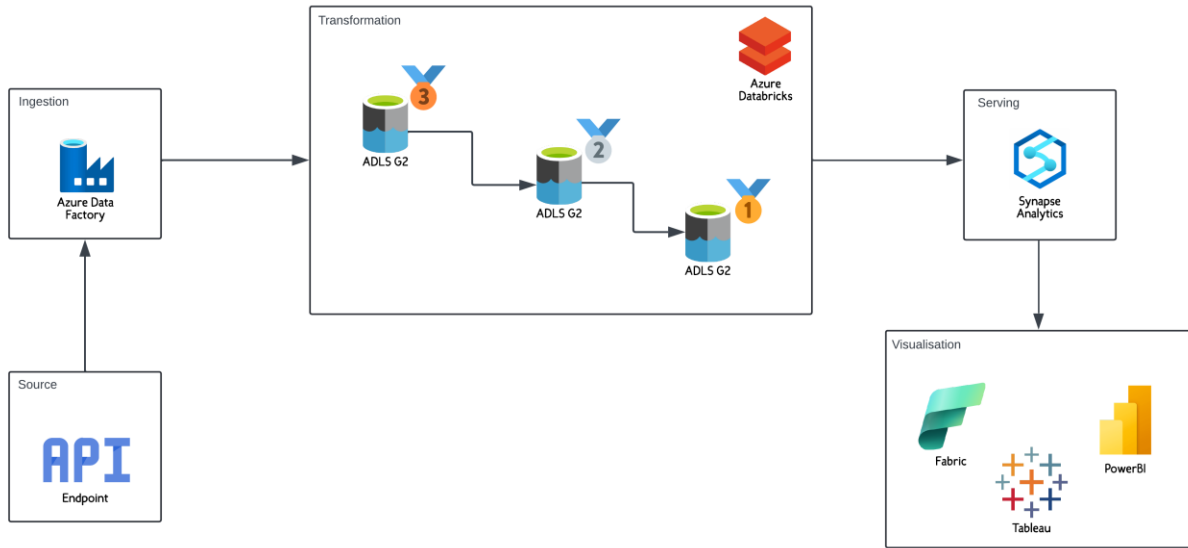
- **Bronze:** Stores the raw, original data as a historical record.
- **Silver:** Contains cleansed and standardized data where nulls and duplicates are removed.
- **Gold:** Store aggregated data that has been modeled into fact and dimension tables and ready for analysis.

Data Loading and Reporting:

Azure Synapse Analytics utilized “Gold” data, which is a refined data used in this analytics solution architecture. Finally, Microsoft Power BI connects to Synapse to display data through interactive dashboards, providing stakeholders with actionable sales and demographic insights.

Automation and Security:

It will run the entire pipeline automatically, so all reports will always be up to date. Azure Key Vault is integrated for storing all enterprise grade password and connection strings so that it maintains high level of security.



Technology Use

This project uses various technologies including on-premises SQL database and a few Microsoft Azure services to build a complete data engineering pipeline. Each service plays a specific role in ensuring that the pipeline is efficient and scalable.

1. On-Premises SQL Database

On-premises SQL database acts as the data source containing the organization data such as the customers and sales data. It represents a standard enterprise system where data is created before being sent to the cloud storage.

2. Azure Data Factory

Azure Data Factory serves as the orchestration layer of the pipeline. It extracts data from the on-premises SQL database and loads it into Azure data lake storage. Besides, it also manages triggers scheduling, workflow automation, and data movement.

3. Azure Data Lake Storage Gen 2

Azure Data Lake Storage Gen 2 works as the central storage system. It stores data in different containers which are the bronze, silver, and gold containers, ensuring scalability and allowing easy access for downstream processing.

4. Azure Databricks

Azure Databricks handles data cleaning, data transformation and data processing. It transforms bronze data into silver data through data cleaning, then transforms silver data into gold data through data transformation.

5. Azure Synapse Analytics

Azure Synapse Analytics performs as the data warehouse and analytics layer. It enables efficient querying and integration of large datasets for reporting and business intelligence.

6. Microsoft Power BI

Microsoft Power BI provides data visualization capabilities. It connects Microsoft Power BI desktop to Azure Synapse Analytics to create interactive dashboards and generate valuable business insights to improve organization's decision-making.

7. Azure Key Vault

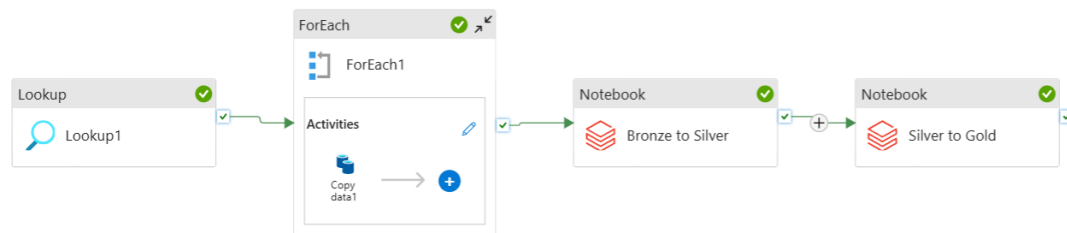
Azure Key Vault ensures security by storing sensitive information such as user credentials, connection strings, and access keys to prevent exposure within the pipeline.

Overall, these technologies work together to form a complete end-of-end data engineering pipeline. Raw data is ingested from the on-premises database using Azure Data Factory and stored in Azure Data Lake Storage Gen 2. Then, the data is transformed into Azure Databricks, analyzed in Azure Synapse Analytics, and finally visualized in Microsoft Power BI. The Azure Key Vault enhances security across all stages.

Data Pipeline Architecture

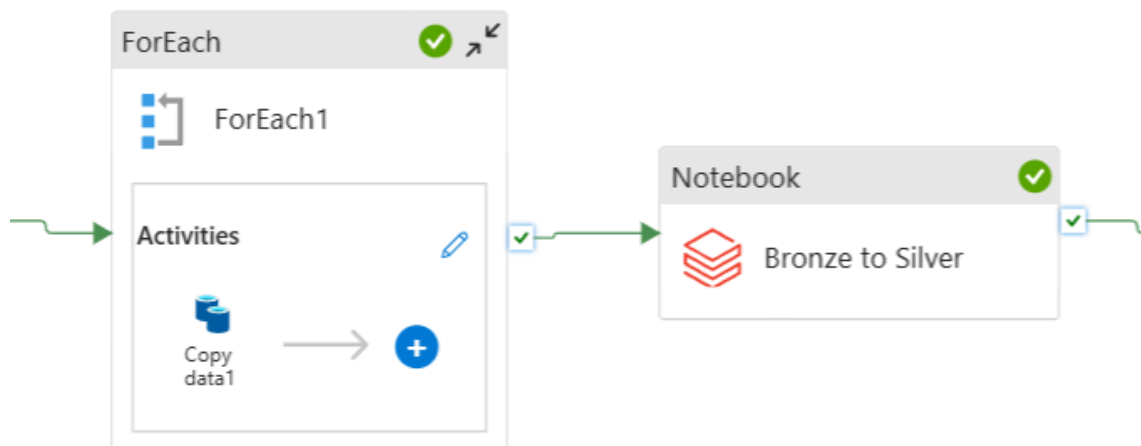
The design of this project is based on the **Medallion Data Architecture** paradigm that logically arranges data within the Azure Data Lake storage Gen 2 into three different layers Bronze (Raw), Silver (Cleansed), and Gold (Curated). The pipeline is a typical batch-processing ETL (Extract, Transform, Load) process. The data flow can be designed as the following main steps:

1. Data Processing (Bronze to Gold Layer)



The pipeline starts with the on-premises SQL Database which contains raw customer and sales data. ADF is the conductor needed to carry out the initial data extraction. ADF runs a "Copy Data" activity using a configured Integration Runtime to make a connection to the Azure cloud in a secure manner by bridging the on-premises network with the cloud. This operation drags the crude, structured data in the SQL tables and deposits it in its original form to the Bronze Container of Azure Data Lake storage (ADLS) Gen 2. This layer is used as the historical record of all raw data.

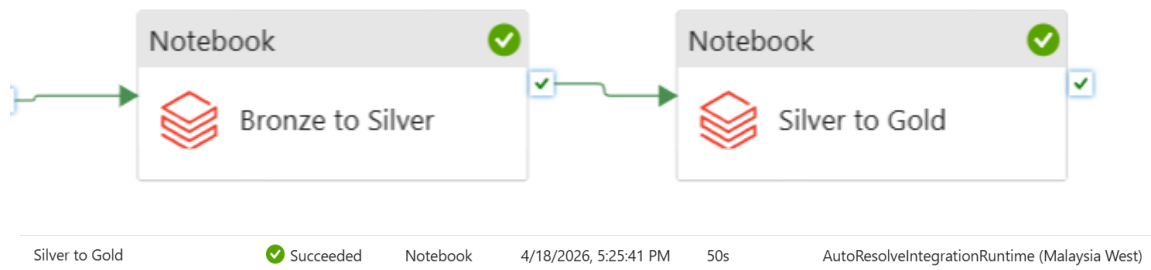
2. Data Cleaning (Bronze to Silver Layer)



Bronze to Silver	✓ Succeeded	Notebook	4/18/2026, 5:24:51 PM	50s	AutoResolveIntegrationRuntime (Malaysia West)
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After the raw data is deposited in the bronze container, the data processing is done by the Azure Databricks. The Databricks notebook is automatically launched, at which point the raw data is loaded, the data undergoes desired rules in data cleaning, and data quality issues like eliminating duplicates, handling nulls, normalizing data formats are resolved. The cleansed and filtered data is then stored to the Silver Container in ADLS Gen 2. This layer offers a solid single source of truth to downstream analytics.

3. Data Transformation (Silver to Gold Layer)



Another Databricks notebook takes the data on the silver layer and makes business-level transformations. At this point, the information is combined, summarized and modelled into an analysis-ready form such as a star scheme of fact and dimension tables. This highly polished data is stored next to the Gold Container in ADLS Gen 2, designed to query and report at high speed.

4. Data Serving and Analytics.

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Pipeline runs

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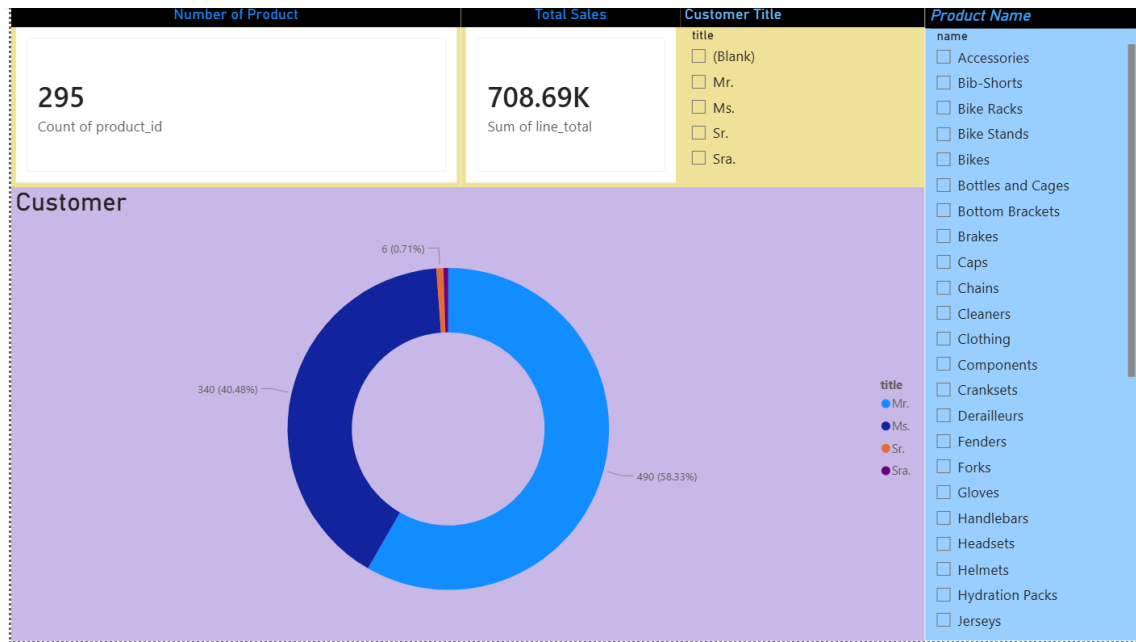
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The data warehousing and serving layer is the Azure Synapse Analytics with the data fully prepared in the gold layer. With Synapse, data engineers and analysts can write complex SQLs on the curated datasets, which connect the data lake to business intelligence tools.

5. Data Visualization



Lastly, Microsoft Power BI is also linked to Azure Synapse Analytics to consume the gold level data. Power BI constructs semantic models to produce interactive dashboards and reports that can give the end-users actionable business insights about customer behaviors and sales performance.

6. Orchestration and Security

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Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Name

Type

Status

databrick-key

✓ Enabled

password

✓ Enabled

username

✓ Enabled

Azure Key Vault is safely incorporated throughout the whole pipeline lifecycle to handle all secrets, including database connection strings, storage account keys, and Databricks tokens. This makes sure that no sensitive data has been hard coded into the pipeline. The Azure Data Factory coordinates the entire process as an end-to-end process and schedules the triggers and controls the data ingestion and the Databricks transformation notebooks dependencies.

Microsoft Power BI Dashboard

The Microsoft Power BI dashboard is the last stage of the data pipeline and the translation of the data that has been highly structured and made analysis-ready in the gold layer into actionable

business insights. Power BI connects to the curated datasets of customers and sales by creating a Direct Query or Import connection to Azure Synapse Analytics, to create an interactive and dynamic reporting interface.

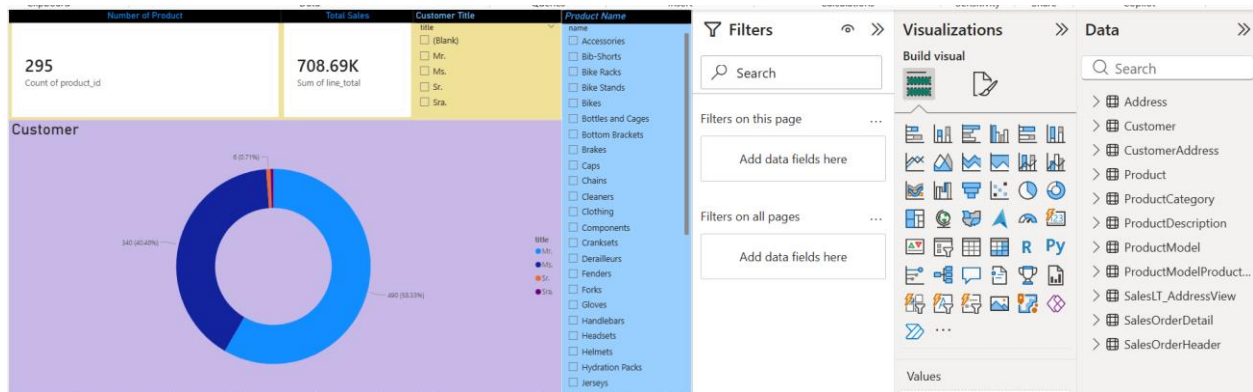


Figure 1: *Overview of the Sales and Customer Power BI Dashboard.*

The dashboard was made to give a detailed overview of the sales performance and customer demographics of the organization. The cards with high-priority metrics, such as Total Sales Revenue, Total Number of Customers, and Total Orders. This gives the decision-makers an up-to-date view of the health of business. These area charts of sales to allow users to see the seasonal trends and when sales are highest. Customer Insights Visualization Pie charts or bar graphs disaggregate the customer base by region, gender or another demographic. Tables or bar charts of the most popular products sold or the most valuable customers and helping to make focused marketing and inventory choices.

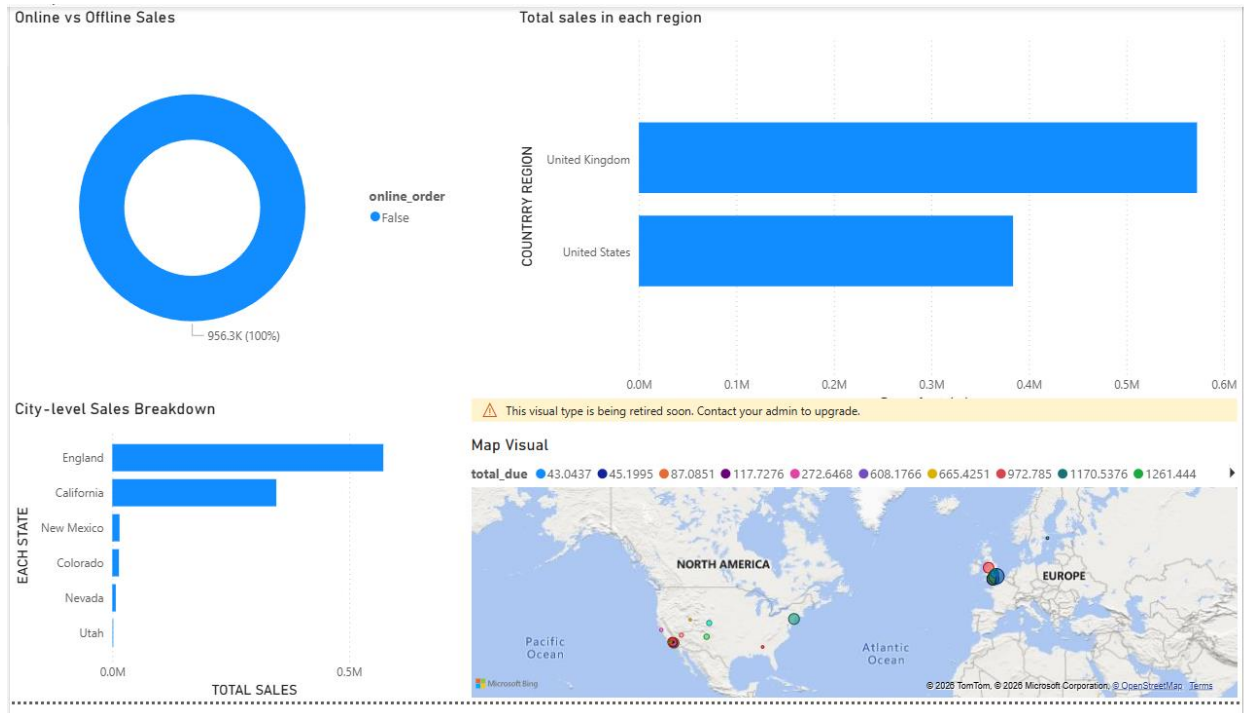


Figure 2: Overview of the combination of 4 graphs in Power BI Dashboard.

1. Online Sales vs Offline Sales.

Chart Type: Donut Chart.

Detailed Description: This chart shows the percentage of revenue generated by various channels of sale. Now, the chart depicts a strong blue circle that reflects 100% of the total sales volume (956.3K). All these sales are under the category of **online_order = False** as per the legend. This means that in the existing set of data, all the revenue is generated through the offline or traditional line of sales, and no online orders have been recorded.

2. Sales in each region.

Chart Type: Clustered Bar Chart.

Detailed Description: These visuals compare sales performance in the various countries at a macro level in descending order. It brings into the limelight that United Kingdom is your best performing market with an overall sale of almost 0.6M (600,000). It performs far better than the United States market which is ranked second with about 0.4M (400,000) in sales.

3. City-level Sales Breakdown

Type of Chart: Clustered Bar Chart.

Detailed Description: The axis labels (EACH STATE) suggest that this chart is a breakdown of sales performance on a state or provincial level, despite the title (City-level). It reveals the regional "heavy weights" within your main markets. The absolute dominant contributor to the sales in the UK is England. In the US market, California stands out by a big margin, yielding significantly more revenue than the other states listed (New Mexico, Colorado, Nevada and Utah).

4. Visual Map

Chart Type: Map

Description: This geographical map will overlay your sales data on actual locations on the map. The bubble position is the position of where orders were made, and the different colors and sizes signify the number of sales (total sales). It graphically supports the evidence presented in the bar charts, making it clear that your customer base and revenue streams are very concentrated in Europe and the West Coast of North America.

Reflection

Elijah She Yu Sheng

The project provided me with great practical knowledge of the development of a modern cloud data pipeline. I took a day to actively debug the Azure test and work through the tutorial video, which was hard, but incredibly rewarding. I went through that intensive process, and it made me understand how to properly and safely connect different Microsoft Azure services to one another, to establish a smooth workflow.

The greatest lesson I learned is that I finally understood how Azure Data Lake and the Medallion Architecture really work in practice. The physical movement of the data in the raw form into the bronze layer, followed by cleaning in the silver layer and final conversion into the analysis-ready gold layer brought into perspective all theoretical ideas. It was such a debugging challenge that finally helped me to conquer my fears of debugging and provided me with a far richer, more practical insight into end-to-end data engineering.

Cheryl Cheong Kah Voon

Through this tutorial, I gained a valuable hands-on experience in creating complete end-to-end data pipeline within the Microsoft Azure ecosystem. I gained a deep understanding of how Azure Data Factory moving data from on-premises sources into cloud native Medallion architecture while orchestrates complex workflow.

Throughout the tutorial, I experience several technical problems including configuring integration runtime, addressing authentication failure with Azure Key Vault, and debugging data refresh delays in the Power BI dashboard. To overcome this problem, I'm working with my teammate and using technologies like Gemini. Overall, this project has overcome the gap between my theoretical knowledge of data engineering and the practical realities of managing scalable cloud infrastructure.

Chua Jia Lin

This project provides me the hands-on experience in building an end-to-end data pipeline using Microsoft Azure services. I learned how to ingest data into data pipelines using Azure Data Factory, store data in Azure Data Lake Storage Gen 2 containers, transform raw data into analysis-ready data through Azure Databricks, perform data analytics using Azure Synapse Analytics, and visualize data using Microsoft Power BI to create interactive dashboards for obtaining meaningful business insights. Moreover, I also learned the settings of Azure Key Vault to ensure security.

During this tutorial, I faced several issues such as configuring Microsoft Azure services, authentication errors, and the dashboard not updated after data pipeline triggers. These challenges are solved with the help of Microsoft Copilot and my teammates to troubleshoot errors, and recheck my steps with the steps demonstrated in the tutorial. While solving these issues, my problem-solving and troubleshooting skills have significantly improved. Overall, this project strengthened my theoretical and practical knowledge about end-to-end data engineering pipelines.